Fairness and Bias in Machine Learning

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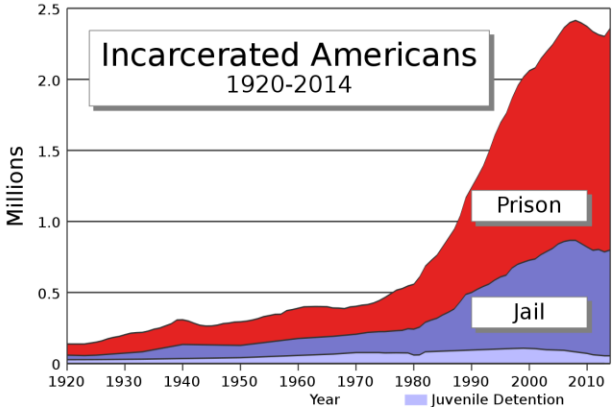
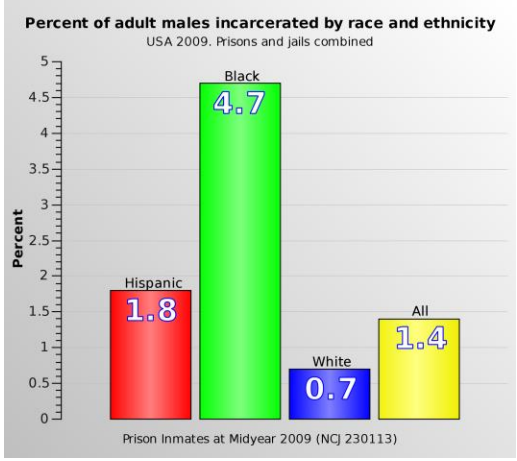
US Criminal Justice System

- The US has the highest incarceration rate in the world: ~0.7%

- Incarceration rates are not uniform across racial groups





Source: Wikipedia

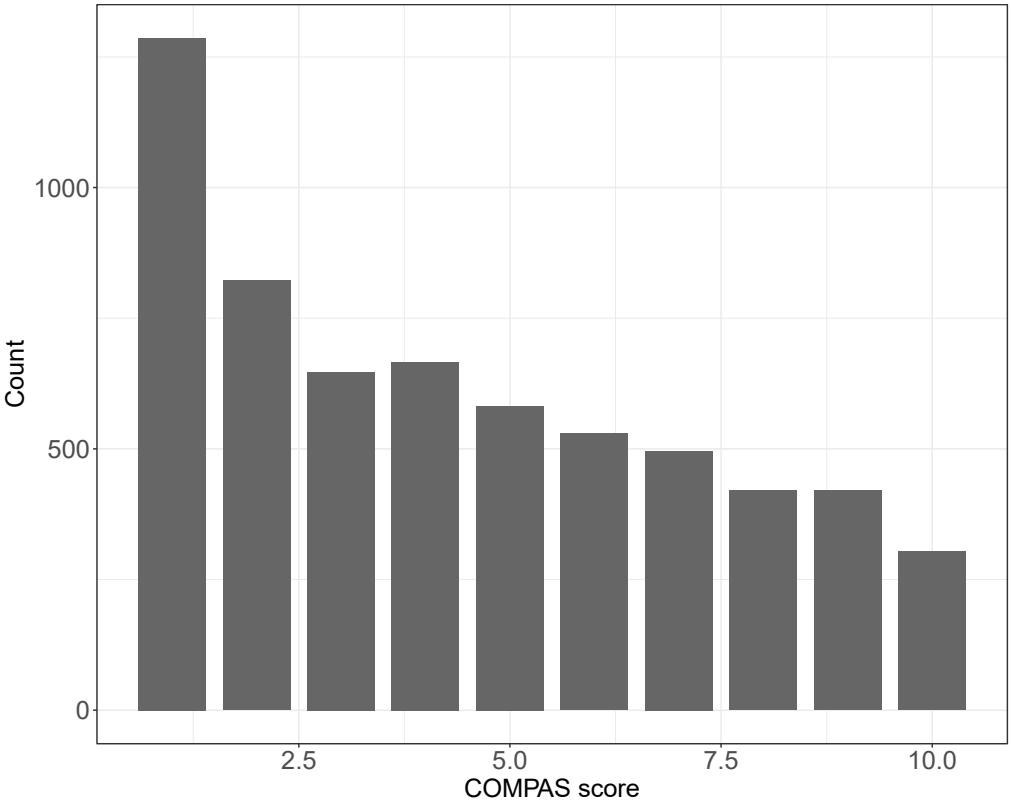
RPI, Northpointe and COMPAS

- Recidivism prediction instruments (RPIs): assessment of recidivism likelihood, guiding pre-trial release & parole decision
- Northpointe: founded in 1989 to develop "software for justice"
- COMPAS (Corrections Offender Management Profiling for Alternative Sanctions): recidivism prediction from 137 questions

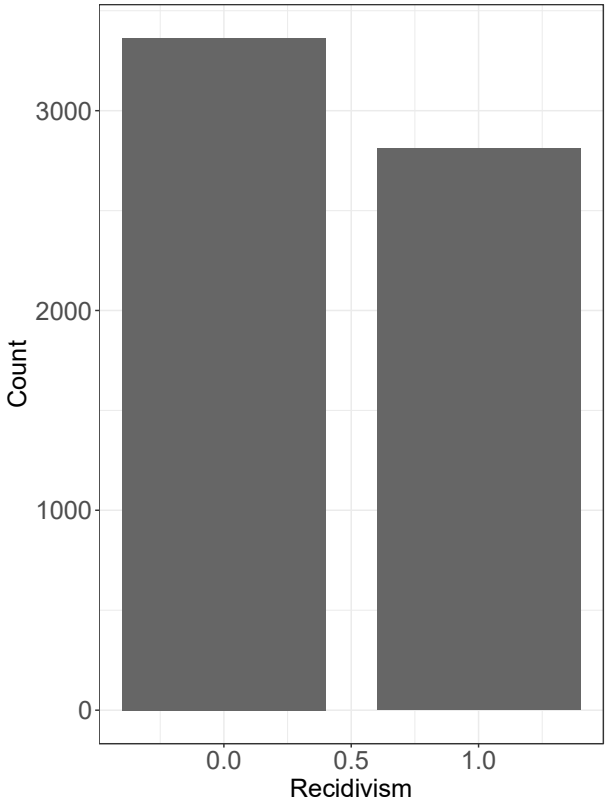
ProPublicas COMPAS dataset

- 7,918 criminal defendants in Broward County, FL in 2013–14; recidivism outcome assess through 2016
- **Outcome Variable**: Risk score Low/High (categorical)
- **Other Variables:**
  - Demographic information: gender, age, race
  - Criminal history: date of offense, number of prior counts, type of previous offense, history of recidivism, felony history, time in prison

Outcome & Predictions

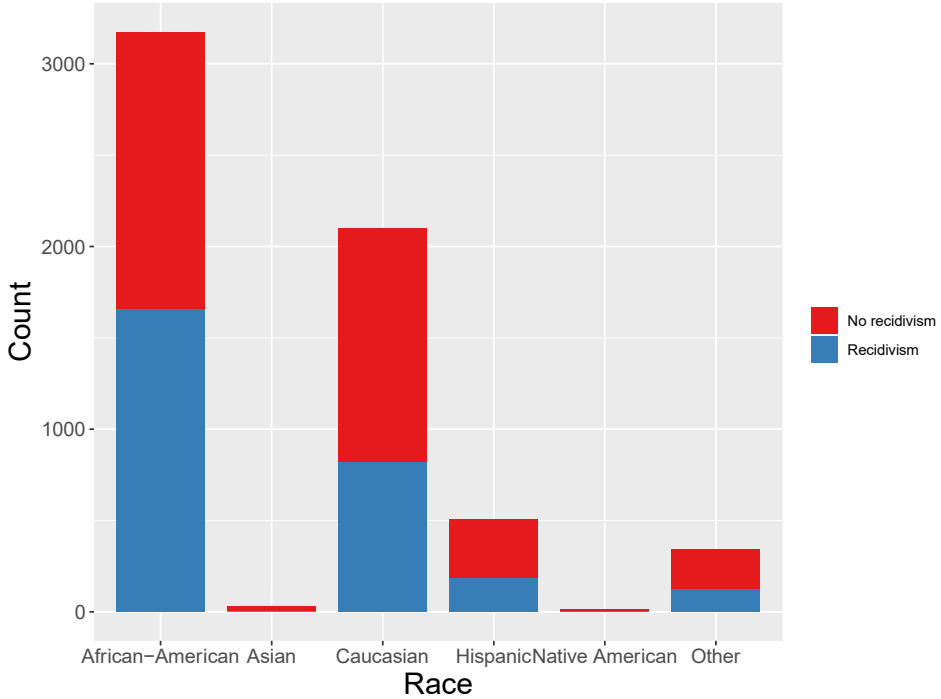


Distribution of COMPAS scores (skewed to the left)

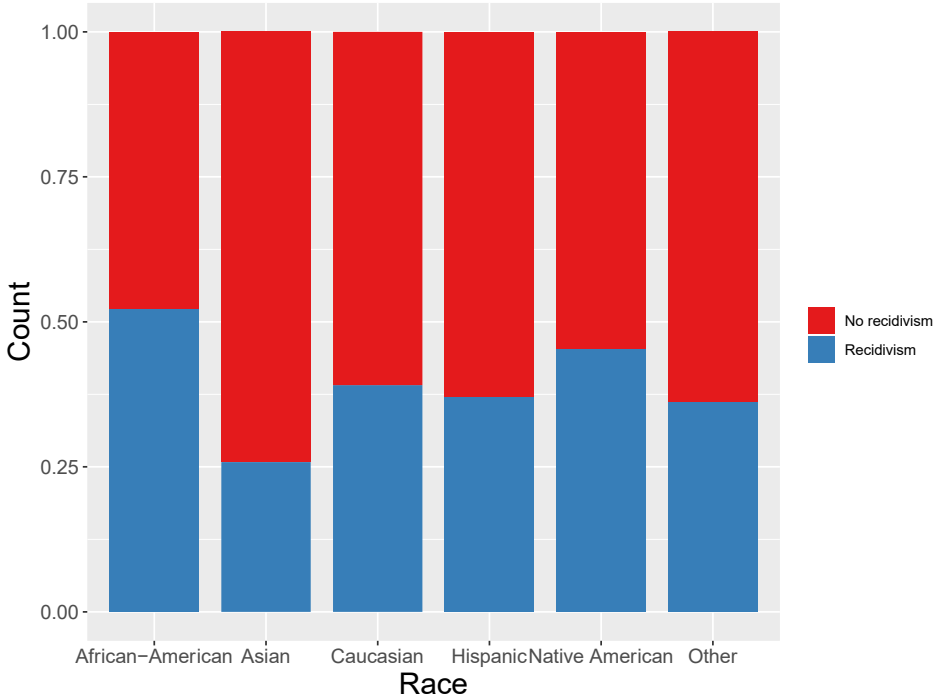


Incidence of recidivism (close to 50%)

Prevalence of Recidivism by Race



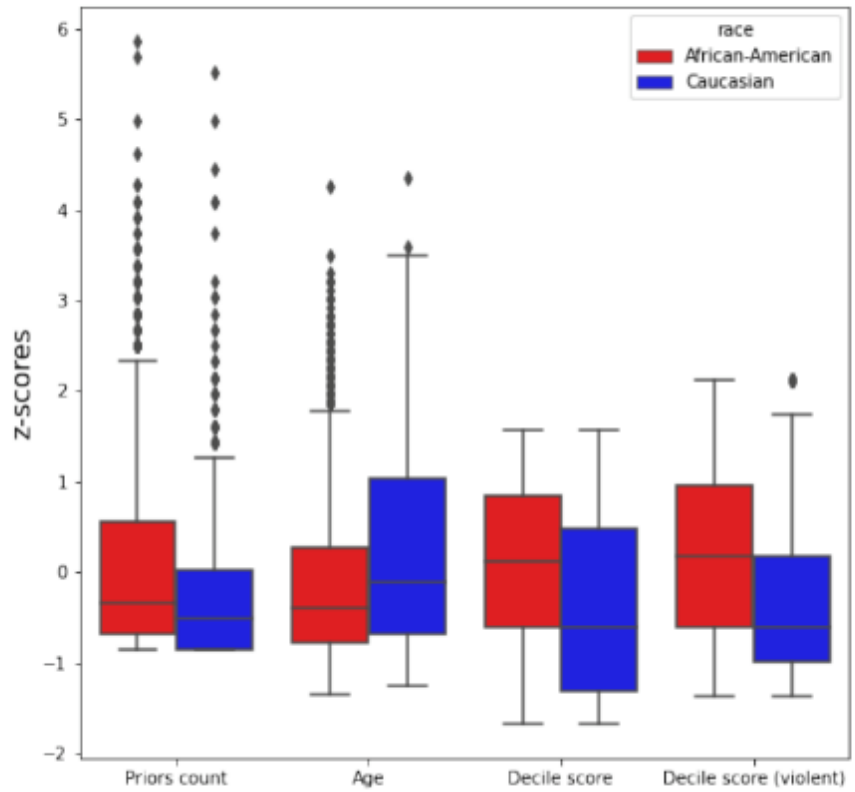
Racial disparities in overall population, and in resulting incidence of recidivism

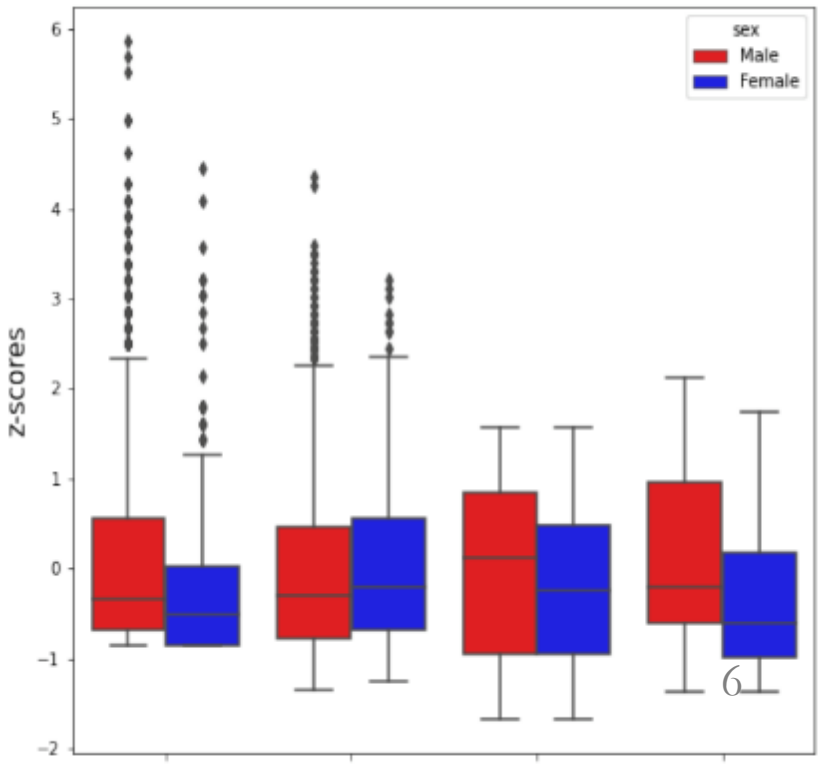


Disparities in the prevalence of recidivism among various sub-populations (here, based on race)

Distributions of covariates by group

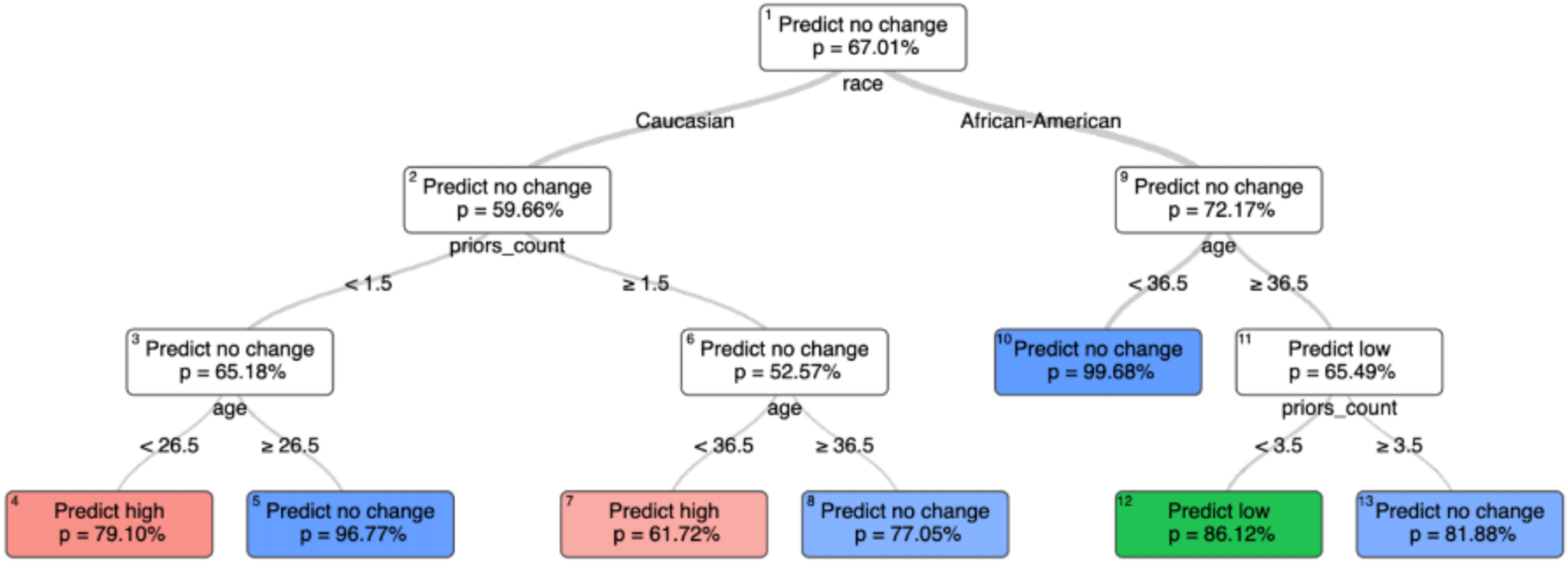
Distributions of age, priors count and COMPAS decile scores for defendants who have been assigned a High risk score by race and gender in the COMPAS dataset ($\alpha_{race} = 0.29$, $\alpha_{gender} = 0.12$).





Actionable insights

Decision tree illustrating rules to achieve demographic parity for race in the COMPAS dataset. Attributes at each of the nodes: race, age, priors_count (number of prior convictions). AUC: 0.72.



Observations

- The logistic regression model identifies criteria (based on Age and prior counts) to select *most qualified* defendants among the black subpopulation
  - (Age $\geq$ 36.5 and prior counts $\leq$ 3) who were *assigned a negative label* and flip their outcome to a *positive label*.
- The model also identifies criteria for the *least qualified* defendants among the white subpopulation
  - (Age < 36.5 & prior counts $\geq$ 1) or (Age < 26.5 & prior counts $\leq$ 1) who were *assigned a positive label* and flipped their outcome to a *negative label*.

Observations

- Leaf nodes that predict label *no change* for the outcome
  - Black subpopulation:
    - (Age < 36.5) – 62.4% of defendants have *low* risk score (*most qualified*).
    - (Age ≥ 36.5 and prior counts ≥ 4)) – 82.6% of defendants have *high* risk score (*least qualified*).

  - White subpopulation:
    - (Age ≥ 26.5 & prior counts ≤ 1) – 91.4% of defendants have *low* risk score (*most qualified*)
    - (Age ≥ 36.5 & prior counts ≥ 1) – 67.7% of defendants have *high* risk score (*least qualified*).

Key takeaways

- Problems involving individuals' choices can be *automated* using Machine Learning algorithms that can *alleviate individuals' biases* and *enhance diversity* without significantly compromising on meritocracy.

- In practice, the price of diversity can be quantified and consequently, demographic parity can be tuned to the level appropriate for the problem at hand.